

STA 5635 HW 4

Palmer Swanson

February 3, 2026

Problem 1

TISP variable selection method ith threshold operator:

$$\Theta(x, \lambda, \eta) = \begin{cases} 0 & \text{if } |x| \leq \lambda \\ \frac{x}{1+\eta} & |x| > \lambda \end{cases}$$

Penalized logistic regression:

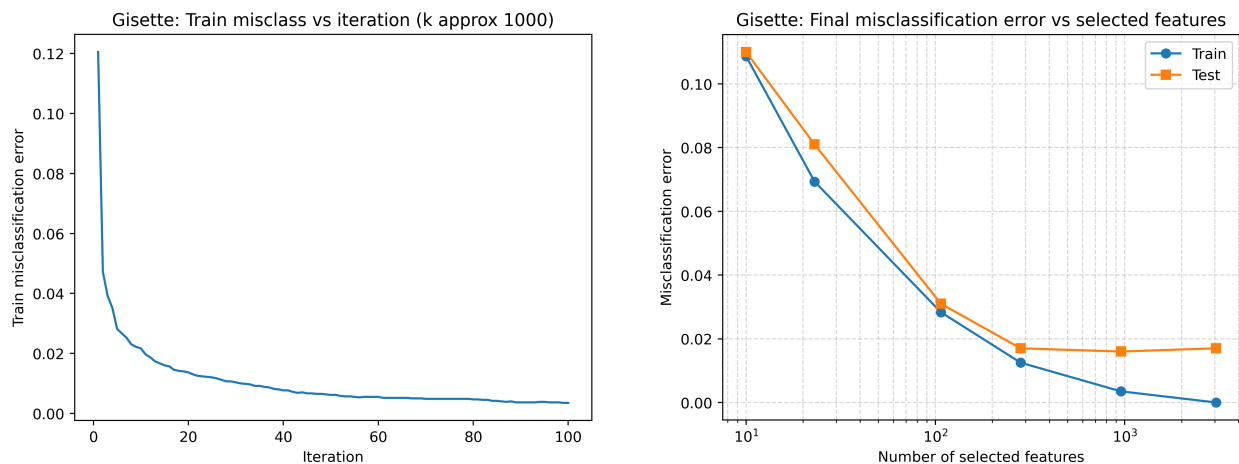
$$L(\omega) = \frac{1}{N} \sum_{j=1}^N \ln(1 + e^{-y_j^* \omega^T x_j}) + \lambda \sum_{i=1}^p p(\omega_i)$$

assuming $y_j^* \in \{-1, 1\}$ and algorithm iteration:

$$\omega^{t+1} = \Theta \left(\omega^t + \eta' X^T \left[\mathbf{y} - \frac{1}{1 + e^{-X \omega^t}} \right], \lambda \right)$$

assuming $y_j \in \{-1, 1\}$.

For $\eta = 0$ and 100 iterations, the first portion of question 1 is shown in Figure 1:



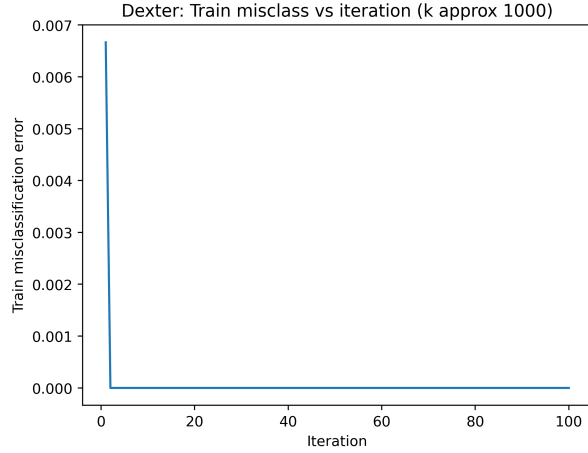
(a) Training misclassification error vs Iteration Number

(b) Train and Test Misclassification error vs Number of Selected Features

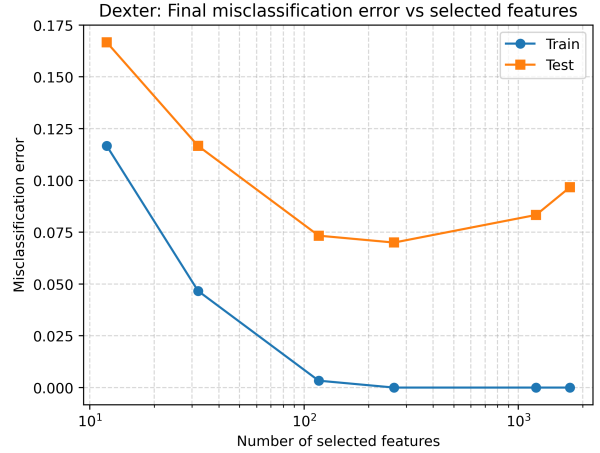
Figure 1: Train Misclassification Error vs. TISP iteration (left) and Train and Test Misclassification Error vs. Number of Selected Features (right). Gisette Dataset.

Second portion of question 1 is shown in Figure 2

The table reporting the train and test misclassification error, the test AUC, corresponding number of features selected, values of λ , and the training time are reported in



(a) Training misclassification error vs Iteration Number



(b) Train and Test Misclassification error vs Number of Selected Features

Figure 2: Train Misclassification Error vs. TISP iteration (left) and Train and Test Misclassification Error vs. Number of Selected Features (right). Dexter Dataset.

Dataset	Target Features	Selected Features	λ	Train Misclass.	Test Misclass.	Test AUC	Train Time (sec)
Gisette	10	10	0.196304	0.108667	0.110000	0.942920	1.812001
Gisette	30	23	0.138489	0.069333	0.081000	0.972464	1.802994
Gisette	100	107	0.086975	0.028333	0.031000	0.993964	1.803976
Gisette	300	282	0.054623	0.012500	0.017000	0.997420	1.803536
Gisette	1000	956	0.019179	0.003500	0.016000	0.998280	2.533000
Gisette	3000	3048	0.004229	0.000000	0.017000	0.997964	1.647994
Dexter	10	12	0.138489	0.116667	0.166667	0.932844	0.351005
Dexter	30	32	0.097701	0.046667	0.116667	0.959911	0.369973
Dexter	100	117	0.068926	0.003333	0.073333	0.975733	0.338997
Dexter	300	263	0.054623	0.000000	0.070000	0.979600	0.369000
Dexter	1000	1210	0.038535	0.000000	0.083333	0.968400	0.548992
Dexter	3000	1739	0.030539	0.000000	0.096667	0.966978	0.352981

Table 1: Train and Test Misclassification Errors, Test AUC, Number of Selected Features, values of λ , and training time, TISP algorithm.

Problem 2

FSA variable selection using logistic loss with shrinkage:

$$L_D(\beta) = \frac{1}{N} \sum_{i=1}^N \ln(1 + e^{-y_i \beta^T \mathbf{x}_j}) + s \|\beta\|_2^2$$

Inverse scheduler with parameter μ :

$$M_i = k + (p - k) \max \left(0, \frac{N^{iter} - 2i}{2i\mu + N^{iter}} \right)$$

For $s = 0.001$, $\mu = 100$, and 100 iterations, the first portion of question 2 is shown in

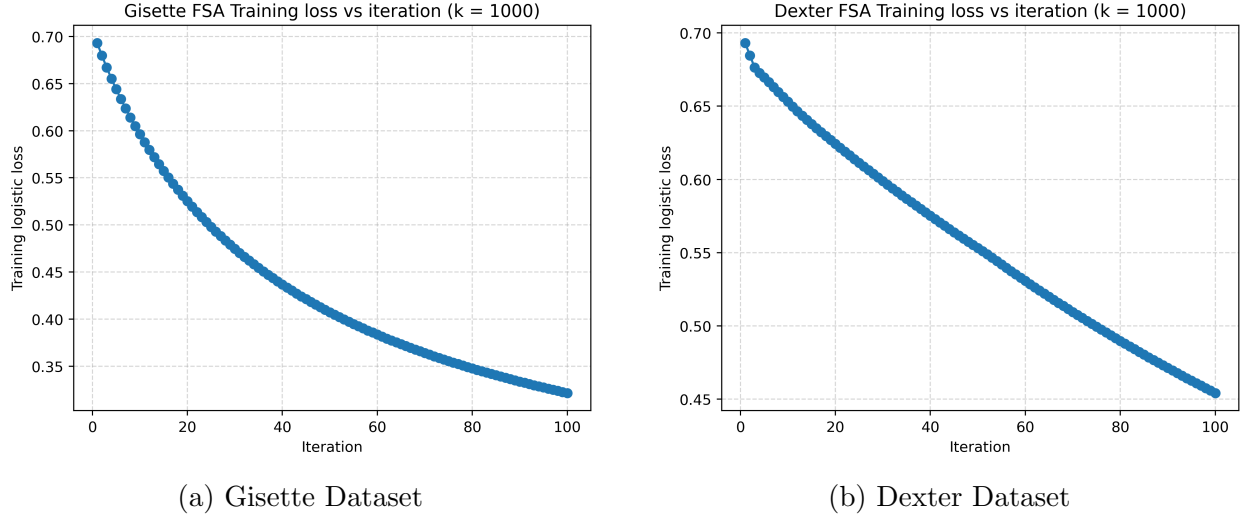


Figure 3: Plot of Training Loss vs. Iteration Number for $k = 1000$. Gisette Dataset (left), Dexter Dataset (right).

At first glance, neither datasets training loss converges, which makes sense. μ is only 100 and we only have 100 iterations; μ is usually 300 and we typically have more iterations.

The table reporting train and test misclassification errors, test AUC, and training time is shown in Table 2:

Dataset	Target k	Selected	Train Miscl.	Test Miscl.	Test AUC	Time (s)
Gisette	10	10	0.139000	0.149000	0.917434	1.673924
Gisette	30	30	0.143333	0.160000	0.928632	1.724052
Gisette	100	100	0.143000	0.156000	0.935600	1.712055
Gisette	300	300	0.112167	0.112000	0.960868	1.706004
Gisette	1000	1000	0.075833	0.072000	0.979972	1.720003
Gisette	3000	3000	0.062500	0.062000	0.984572	1.708005
Dexter	10	10	0.280000	0.290000	0.917489	0.412967
Dexter	30	30	0.173333	0.193333	0.936178	0.436001
Dexter	100	100	0.056667	0.106667	0.972356	0.415999
Dexter	300	300	0.013333	0.090000	0.975378	0.415000
Dexter	1000	1000	0.003333	0.080000	0.972667	0.418003
Dexter	3000	3000	0.000000	0.123333	0.938000	0.414001

Table 2: Train and Test Misclassification Errors, Test AUC, and training time, FSA algorithm.